

1. **Random Experiment:** If in each trail an experiment conducted under identical conditions, the outcome is not unique, but may be one of the possible outcomes, then such an experiment is called a *random experiment*.
Example: Throwing a die, a pack of cards.
2. **Outcome:** The result of a random experiment will be called an *outcome*.
3. **Trial:** Any particular performance of a random experiment is called a ***trial*** and outcome or combination of outcomes are termed as ***events***.
Example: Tossing of a coin: {H, T}
4. **Exhaustive events or cases:** The total number of possible outcomes of a random experiments is known as the exhaustive events or cases.
5. **Favourable Events or Cases:** The number of cases favourable to an event in a trail is the number of outcomes which entail the happening of the event.
Example: In drawing a card from a pack of cards the number of cases favourable to drawing of an ace is 4,
for drawing a spade card is 13 and
for drawing a red card is 26.

6. ***Mutually Exclusive events:*** Events are said to be mutually exclusive or incompatible if the happening of any one of them precludes the happening of all the others, *i.e.*, no two or more of them can happen simultaneously in the same trail.
7. ***Equally likely events:*** Outcomes of trail are said to be equally likely if taking into consideration all the relevant evidences, there is no reason to expect one in preference to the others.
8. ***Independent Events:*** Several events are said to be independent if the happening (non-happening) of an event is not affected by the supplementary knowledge concerning the occurrence of any number of remaining events.

Statistical Definition:

If a random experiment or trail results in ' n ' exhaustive, mutually exclusive and equally likely outcomes(or cases) , out of which ' m ' are favourable to the occurrence of an event E , then the probability ' p ' of occurrence of E , denoted by $P(E)$, is given by:

$$p = P(E) = \frac{\text{Number of favourable cases}}{\text{Total number of exhaustive cases}} = \frac{m}{n}$$

(i) Since $m \geq 0, n \geq 0$ and $m \leq n$

$$P(E) \geq 0 \text{ and } P(E) \leq 1 \implies 0 \leq P(E) \leq 1$$

(ii) $P(E) + P(\bar{E}) = 1$

Axiomatic Probability

Definition: $P(A)$ is the probability function defined on a σ –field $\mathbf{B}$ of events if the following properties or axioms hold.

1. For each $A \in \mathbf{B}$, $P(A)$ is defined, is real and $P(A) \geq 0$. (Axiom of non-negativity)
2. $P(S) = 1$ (Axiom of certainty)
3. If $\{A_n\}$ is any finite or infinite sequence of disjoint events in $\mathbf{B}$, then

$$P(\cup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i) \quad \text{(Axiom of additivity)}$$

4. The set function $\mathbf{P}$ defined on σ –field $\mathbf{B}$, taking its values in the real line and satisfying the above three axioms is called the *probability measure*.

Algebra of Events:

For events A, B, C

(i) $A \cup B = \{\omega \in S: \omega \in A \text{ or } \omega \in B\}$

(ii) $A \cap B = \{\omega \in S: \omega \in A \text{ and } \omega \in B\}$

(iii) $\bar{A} = \{\omega \in S: \omega \notin A\}$

(iv) $A - B = \{\omega \in S: \omega \in A \text{ but } \omega \notin B\}$

(v) $A \subset B$ for every $\omega \in A, \omega \in B$

(vi) $A = B$ if and only if A and B have same elements, *i. e.*, $A \subset B$ and $B \subset A$

(vii) A and B disjoint events (mutually exclusive) $\Rightarrow A \cap B = \Phi$ (*null set*)

(viii) $A \cup B$ can be denoted by $A + B$ if A and B are disjoint.

(ix) $A \Delta B$ denotes those ω belonging to exactly one of A and B, *i.e.*,

$$A \Delta B = \bar{A} B \cup A \bar{B} = \bar{A} B + A \bar{B} \quad (\text{disjoint events})$$

Theorems of Probability

1) Addition Theorem: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

2) $P(\cup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i) - \sum \sum P(A_i \cap A_j) + \dots + (-1)^{n-1} P(A_1 \cap A_2 \cap \dots \cap A_n)$

Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

If the events A and B are independent, $P(A|B) = P(A)$

$$P(B|A) = P(B)$$

Multiplication theorem of Probability for Independent events:

$$P(A \cap B) = P(A) \cdot P(B)$$

For n events, $P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2|A_1) P(A_3|A_1 \cap A_2) \dots \times$
 $P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$

Bayes Theorem of Probability:

Theorem: If $E_1, E_2, \dots, E_n$ are mutually disjoint events with $P(E_i) \neq 0$, ($i = 1, 2, \dots, n$), then for any arbitrary event A which is a subset of $\cup_{i=1}^n (E_i)$ such that $P(A) > 0$, we have

$$P(E_i|A) = \frac{P(E_i) P(A|E_i)}{\sum_{i=1}^n P(E_i) P(A|E_i)} ; i = 1, 2, \dots, n$$

A real number X connected with the outcome of a random experiment E .

For example, if E consists of two tosses the random variable which is the number of **heads** (0,1 or 2).

Outcome	:	<i>HH</i>	<i>HT</i>	<i>TH</i>	<i>TT</i>
Value of X	:	2	1	1	0

Thus to each outcome ω , there corresponds a real number $X(\omega)$.

Random Variable

Let S be the sample space associated with a given random experiment $\mathbf{E}$.

A real-valued function defined on S and taking values in $\mathbf{R} (-\infty, \infty)$ is called a ***one-dimensional random variable***.

If the function values are ordered pairs of real numbers (*i.e.*, vectors in two-space), the function is said to be a ***two-dimensional random variable***.

More generally, an ***n***-dimensional random variable is simply a function whose domain is S and whose range is a collection of ***n***-tuples of real numbers (vectors in ***n***-space).

Random variable:

it is defined as a real valued function on the sample space taking values on the real line.

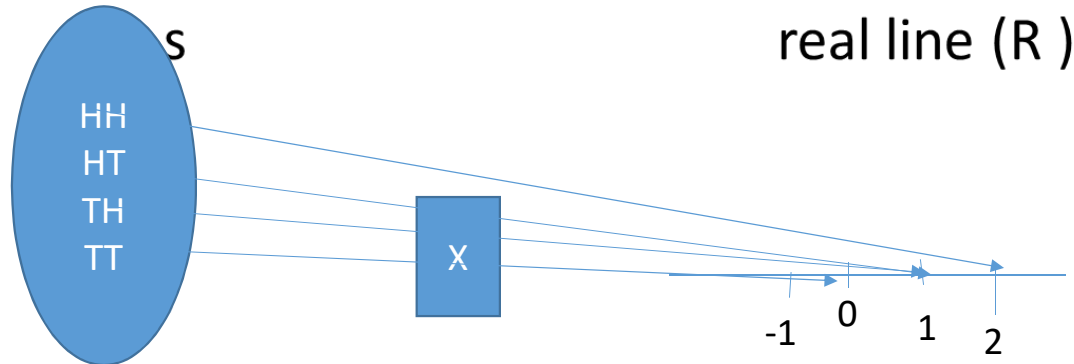
$$X: \mathcal{S} \rightarrow \mathcal{R}$$

Ex: If E consists of two tosses of a coin

consider the random variable X which is the number of heads (0,1,2)

outcomes : HH HT TH TT

No. of Heads : 2 1 1 0



$P(X = a) \rightarrow$ the probability of the event X assuming the value 'a'.

$P(a < X < b) \rightarrow$ the probability of the event X
assuming values between a and b .

$P(X \leq c) \rightarrow$ the probability of the event $X \leq c$.

$P(X > c) \rightarrow$ the probability of the event $X > c$.

Discrete Random Variable

Let $\mathbf{X}$ be a finite random variable on a sample space $\mathbf{S}$, that is, $\mathbf{X}$ assigns only finite number or countably infinite number of values to $\mathbf{S}$.

$$\text{Say, } R_X = \{x_1, x_2, \dots, x_n, \dots, \infty\}$$

Example: marks obtained in a test, number of accidents per month, number of telephone calls per unit time, number of successes in n trials and so on.

Then, $\mathbf{X}$ induces a function $\mathbf{p}$ which assigns probabilities to the points in $\mathbf{R}_X$ as follows:

$$p(x_i) = p_X(x_i) = P(X = x_i) = P\{s \in S : X(s) = x_i\} \text{ for } i = 1, 2, \dots, n$$

Probability Mass function: If X is a discrete random variable with distinct values $x_1, x_2, \dots, x_n$ then the function $\mathbf{p}_X(\mathbf{x})$ is defined as :

$$p_X(x_i) = \begin{cases} P(X = x_i) = p_i, & \text{if } x = x_i \\ 0, & \text{if } x \neq x_i ; i = 1, 2, \dots, n \end{cases}$$

Is called the **probability mass function** of random variable $\mathbf{X}$.

Let X be a discrete R.V. with possible out comes $x_1, x_2, x_3, \dots x_n$.

The probability of each outcome x_i is $P(X = x_i) = p(x_i) \quad \forall i$

X	x_1	x_2	x_3	..	..	x_n
P(X)	$p(x_1)$	$p(x_2)$	$p(x_3)$	..	..	$p(x_n)$

Ex: when a coin is tossed twice.

$S = \{TT, HT, TH, HH\}$

If X denote the number of heads
the probability distribution is given by

$X = x_i$	0	1	2
$p(X = x_i)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

No head case

one head case

two heads case

probability values

The numbers $p(x_i); i = 1, 2, \dots$ must satisfy the following conditions:

$$(i) \quad p(x_i) \geq 0 \quad \forall \quad i$$

$$(ii) \quad \sum_{i=1}^n p(x_i) = 1$$

Distribution Function

Definition: Let $\mathbf{X}$ be a random variable. The function $\mathbf{F}$ defined for all real $\mathbf{x}$ by

$$F_X(x) = F(x) = P(X \leq x) = P\{\omega : X(\omega) \leq x\}, -\infty < x < \infty$$

is called the *distribution function* or *cumulative distribution function* of r.v. ($\mathbf{X}$).

The domain of the distribution function is $(-\infty, \infty)$ and its range is $[0,1]$

Properties:

If x is a real number, the set of all ω in S such that $X(\omega) = x$ is, denoted by $X = x$.

$$1. \quad P(X = x) = P(\omega : X(\omega) = x)$$

$$2. \quad P(X \leq a) = P\{\omega : X(\omega) \in (-\infty, a] \}$$

$$3. \quad P(a < X \leq b) = P\{\omega : X(\omega) \in (a, b] \} = F(b) - F(a)$$

$$P(a \leq X \leq b) = P(X = a) + [F(b) - F(a)]$$

$$P(a \leq X < b) = F(b) - F(a) - P(X = b) + P(X = a)$$

$$P(a < X < b) = F(b) - F(a) - P(X = b)$$

$$4. \quad P(X = a \text{ or } X = b) = P\{(X = a) \cup (X = b)\}$$

$$5. \quad P(X = a \text{ and } X = b) = P\{(X = a) \cap (X = b)\}$$

Properties

If F is the *distribution function* of the r.v. X and if $a < b$, then

$$6. \quad P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a)$$

$$7. \quad \text{(i) } 0 \leq F(x) \leq 1$$

$$\text{(ii) } F(x) \leq F(y) \text{ if } x < y$$

$$8. \quad F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0 \quad \text{and}$$

$$F(\infty) = \lim_{x \rightarrow \infty} F(x) = 1$$

Discrete Distribution Function:

A countable number of points $x_1, x_2, x_3, \dots, x_n$

$$p(x_i) \geq 0 \quad \forall i,$$

$$\sum_{i=1}^n p(x_i) = 1 \quad \text{such that}$$

$$F(x) = \sum_{i: x_i \leq x} p_i = \sum_{x_i} P(X = x_i)$$

If x_i is just integer i , so that $P(X = i) = p_i$; $i=1, 2, 3, \dots$. Then **$F(x)$** is a “**step function**” having jump **p** at i and being constant between each pair of integers.

Properties:

1. $p(x_j) = P(X = x_j) = F(x_j) - F(x_{j-1})$, where F is the *d.f.* of X .

Example 1: A random variable X has the following probability function:

Values of X , x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

- (i) Find k , (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$
- (iii) Determine the distribution function of X .

Example 2:

Two dice are rolled. Let X denote the random variable which counts the total number of points on the upturned faces, Construct a table giving the non-zero values of the *p.m.f* and draw the probability chart. Also find the distribution function of X .

Continuous Random Variable

Definition: A random variable X is said to be continuous if it can take all possible values between certain limits.

A continuous random variable is a r.v. that can be measured to any desired degree of accuracy.

Example: Age, height, weight etc.

Probability Density Function (p.d.f)

Definition: Consider the small interval $(x, x + dx)$ of length dx round the point x . Let $f(x)$ be any continuous function of x so that $f(x)dx$ represents the probability that X falls in the infinitesimal interval $(x, x + dx)$.

$$P(x \leq X \leq x + dx) = f_X(x)dx$$

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{P(x \leq X \leq x + \delta x)}{\delta x}$$

The probability for a variate value to lie in the interval dx is $f(x)dx$ and hence the probability for a variate value to fall in the interval $[\alpha, \beta]$ is:

$$P(\alpha \leq X \leq \beta) = \int_{\alpha}^{\beta} f(x)dx$$

Continuous Distribution Function

Definition: If X is continuous *r.v.* with the *p.d.f.* $f(x)$, then the function

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt \ ; \ -\infty < x < \infty$$

is called the distribution function (*d.f.*) or sometimes the cumulative distribution function (*c.d.f.*) of the random variable X .

Properties of *p. d. f.* :

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

(iii) The probability $P(E)$ given by: $\int_E f(x)dx$ is well defined for any event E .

(iv) $P(X = c) = 0, \forall c$

(v) $P(\alpha \leq X \leq \beta) = P(\alpha \leq X < \beta) = P(\alpha < X \leq \beta) = P(\alpha < X < \beta)$

Properties: 1. $0 \leq F(x) \leq 1$; $-\infty < x < \infty$

2. $F'(x) = \frac{d}{dx} F(x) = f(x) \geq 0$

$\Rightarrow F(x)$ is non-decreasing function of x .

3. $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} \int_{-\infty}^x f(x) dx = \int_{-\infty}^{-\infty} f(x) dx = 0$

$$F(+\infty) = \lim_{x \rightarrow \infty} F(x) = \lim_{x \rightarrow \infty} \int_{-\infty}^x f(x) dx = \int_{-\infty}^{\infty} f(x) dx = 1$$

4. $F(x)$ is continuous function of x on the right.

5. The discontinuities of $F(x)$ are at the most countable.

Example 3:

The diameter of an electric cable, say X , is assumed to be a continuous *r.v.* with *p.d.f*: $f(x) = 6x(1 - x)$; $0 \leq x \leq 1$

- (i) Check that $f(x)$ is p.d.f,
- (ii) Determine a number b such that $P(X < b) = P(X > b)$.

Example 4:

Let X be a continuous random variable with *p.d.f.* :

$$f(x) = \begin{cases} ax & ; 0 \leq x \leq 1 \\ a & ; 1 \leq x \leq 2 \\ -ax + 3a & ; 2 \leq x \leq 3 \\ 0 & ; \text{elsewhere} \end{cases}$$

- (i) Determine the constant ***a***.
- (ii) Compute $P(X \leq 1.5)$.

Median: In case of continuous distribution, median is the point which divides the total area two equal parts. Thus if M is the median, then

$$\int_a^M f(x) dx = \int_M^b f(x) dx = \frac{1}{2}$$

Thus solving $\int_a^M f(x) dx = \frac{1}{2}$ or $\int_M^b f(x) dx = \frac{1}{2}$ for M , we get the median value.

Mode: Mode is the value of x for which $f(x)$ is maximum. Mode is thus solution of

$$f''(x) = 0 \text{ and } f''(x) < 0, \text{ provided it lies in } [a, b].$$

Example 5:

Suppose that the life in hours of a certain part of radio tube is a continuous random variable X with the *p.d.f.* given by:

$$f(x) = \begin{cases} \frac{100}{x^2}, & x \geq 100 \\ 0, & \text{elsewhere} \end{cases}$$

- (i) What is the probability that all of three such tubes in a given radio set will have to be replaced during the first 150 hours of operation?
- (ii) What is the probability that none of three of the original tubes will have to be replaced during that first 150 hours of operation?
- (iii) What is the probability that a tube will last less than 200 hours if it is known that the tube is still function after 150 hours of service?

Example 6:

A petrol pump is supplied with petrol once a day. If its daily volume of sales (X) in thousands of litres is distributed by:

$$f(x) = 5(1 - x)^4 ; \quad 0 \leq x \leq 1$$

What must be the capacity of its tank in order that the probability that its supply will be exhausted in a given day shall be 0.01?